

Contents

Hybrid Mortgage Calculator	1
Setup	1
Direct calculation	1
Natural-language mode	1
Amortization	1
Scope and disclaimer	2

Hybrid Mortgage Calculator

Chapter 31 implements a fixed-rate, monthly-payment mortgage calculator with a deterministic financial core and an optional natural-language boundary.

Setup

```
uv sync --extra test
uv run pytest
```

The test suite is offline and does not require an API key, model server, or network access.

Direct calculation

Provide exactly three primary quantities. Annual rates are percentages by default; terms in years are converted to monthly payments.

```
uv run mortgage calculate \
  --principal 500000 \
  --rate 6.5 \
  --term-years 30
```

Use `--format json` for a machine-readable response. Decimal values are serialized as strings and integer counts as JSON integers.

Natural-language mode

The default adapter is the deterministic offline mock:

```
uv run mortgage ask --adapter mock \
  'What is the payment on $500,000 at 6.5% for 30 years?'
```

The adapter interprets language and calls the same calculator tool used by direct mode. It never performs arithmetic itself. A real local Ollama adapter is available as an opt-in path:

```
ollama pull llama3.2
uv run mortgage ask --adapter real --model llama3.2 \
  'What is the payment on $500,000 at 6.5% for 30 years?'
```

OLLAMA_HOST defaults to `http://localhost:11434` and can be overridden with `--host`; OLLAMA_MODEL supplies the default model when `--model` is omitted. Ollama is used only for interpretation and explanation. The deterministic calculator tool remains authoritative for every financial number. Real-model failures return exit code 5 and never fall back to model-generated arithmetic.

Amortization

Request a bounded schedule with:

```
uv run mortgage amortize \  
  --principal 500000 \  
  --rate 6.5 \  
  --term-years 30 \  
  --format json
```

Schedules are limited to 1,200 payment periods. The final payoff row is adjusted when needed to close the balance while preserving the payment identity.

Scope and disclaimer

The calculator supports fixed-rate loans with monthly payments and principal-and-interest calculations only. It does not estimate taxes, insurance, HOA fees, lender fees, adjustable-rate loans, or lender-specific quotes.

Every response includes:

- Estimate for principal and interest only; not a lender-specific quote or financial advice.